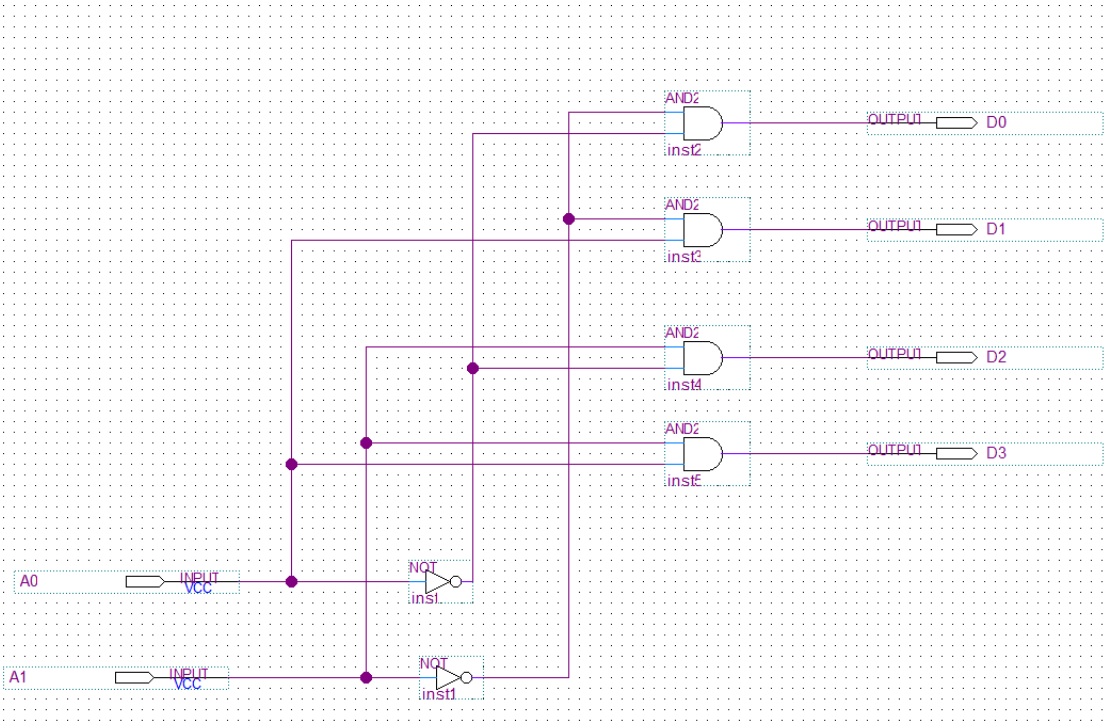


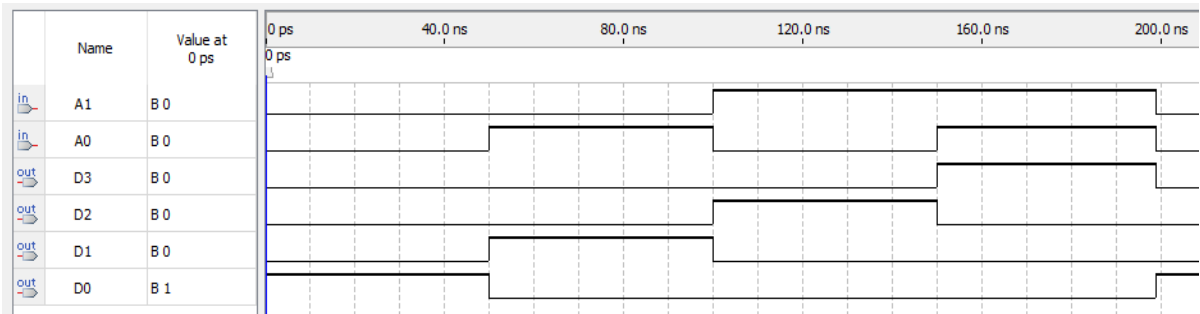
BASIC CIRCUITS

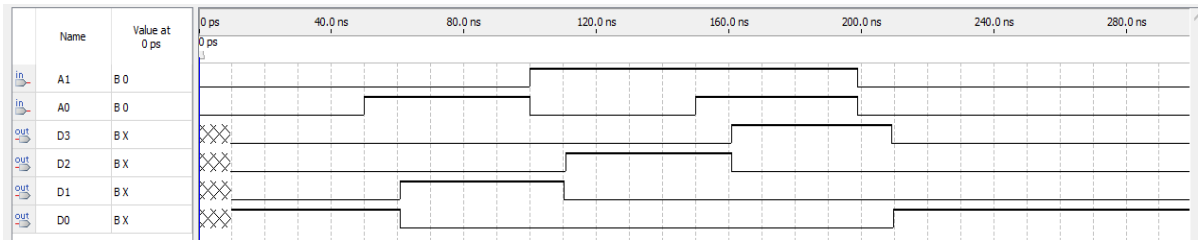
Decoder 2 to 4

Circuit:



The Decoder 2 to 4 functions correctly as it can be seen bellow:

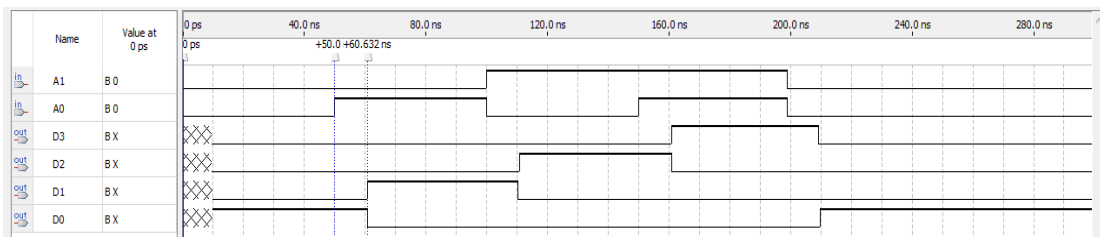




The biggest delay, according to the static analysis, is 10.857ns:

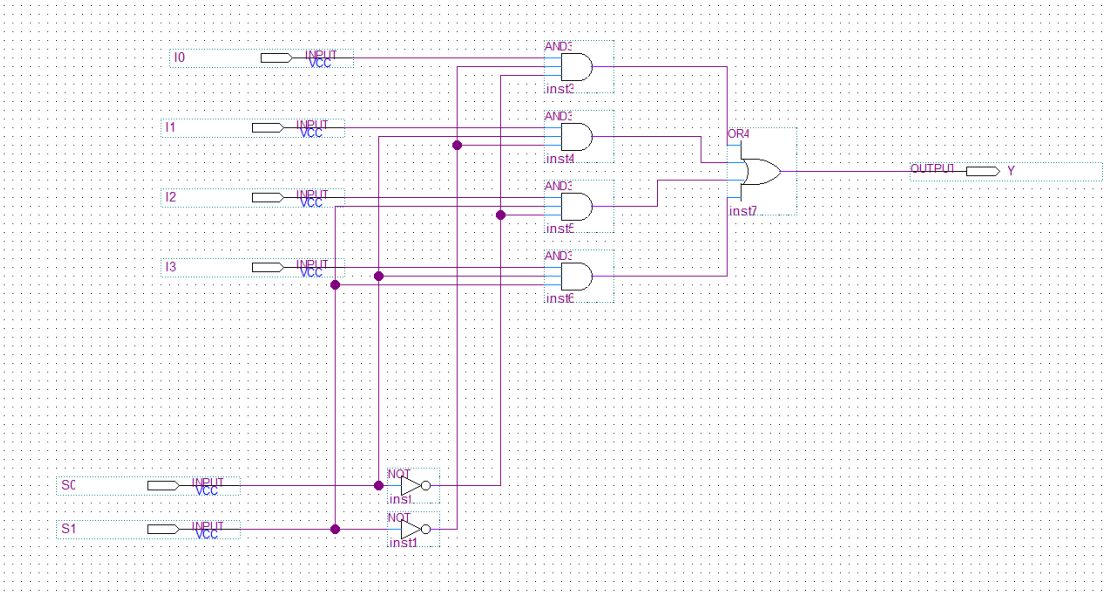
	Input Port	Output Port	RR	RF	FR	FF
1	A0	D0		10.848	10.848	
2	A0	D1	10.857			10.857
3	A0	D2		10.826	10.826	
4	A0	D3	10.833			10.833
5	A1	D0		10.322	10.322	
6	A1	D1		10.363	10.363	
7	A1	D2	10.332			10.332
8	A1	D3	10.334			10.334

The biggest delay, according to the waveform, is 10.632 ns:

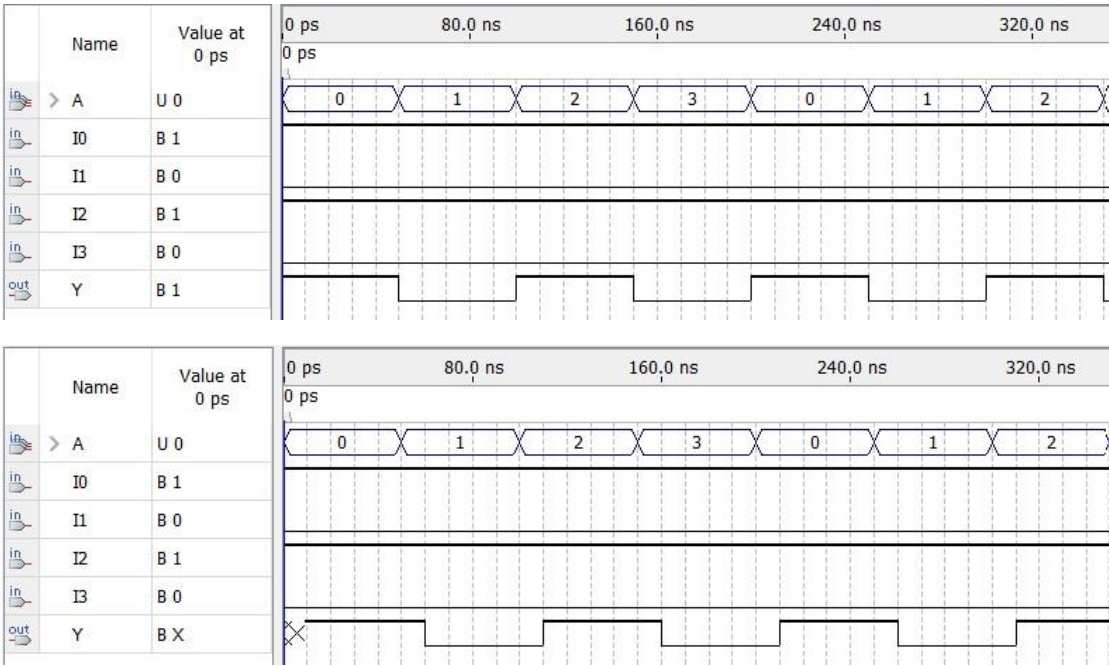


Multiplexer 4 to 1

Circuit:



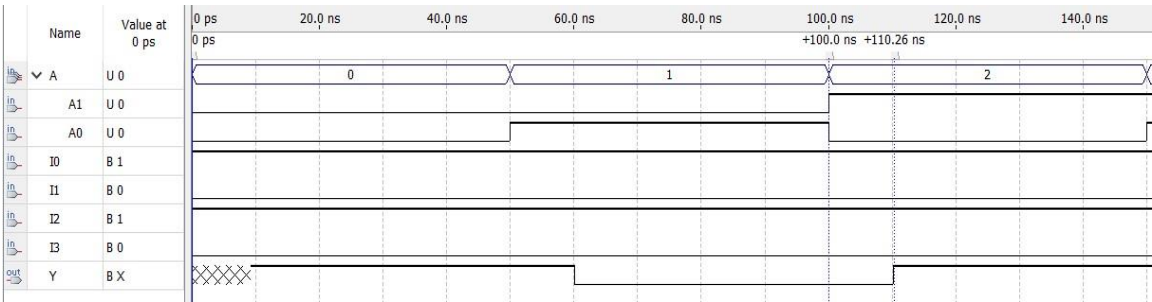
The Multiplexer 1 to 4 functions correctly as it can be seen below:



The biggest delay, according to the static analysis, is 10.156ns:

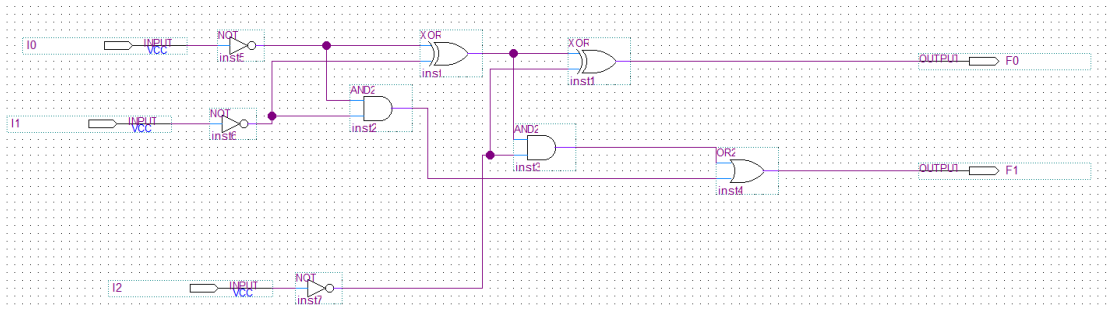
	Input Port	Output Port	RR	RF	FR	FF
1	A0	Y	10.156	10.156	10.156	10.156
2	A1	Y	6.046	6.046	6.046	6.046
3	I0	Y	9.964	9.964	9.964	9.964
4	I1	Y	10.094	10.094	10.094	10.094
5	I2	Y	5.500			5.500
6	I3	Y	9.481			9.481

The biggest delay, according to the waveform, is 10.26 ns:

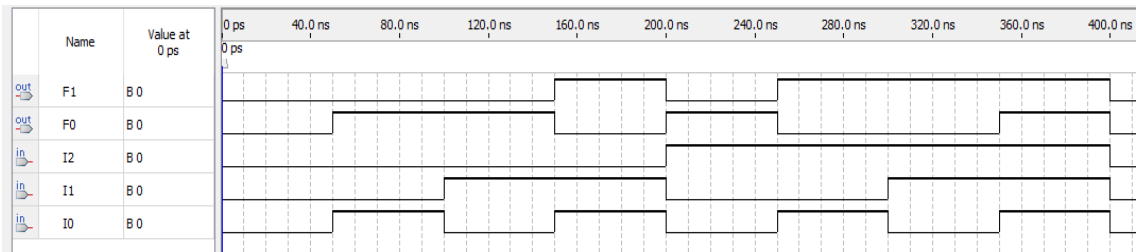


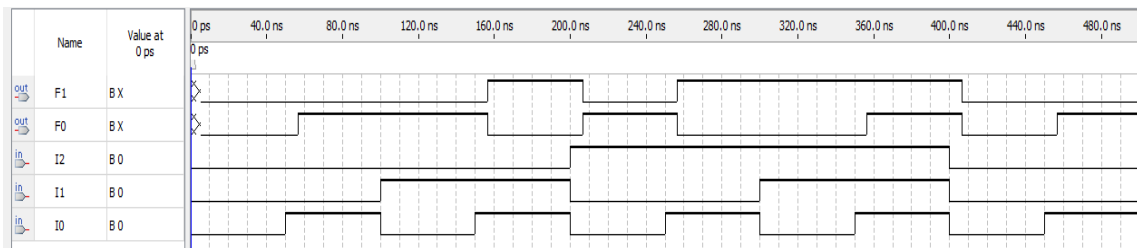
Full Adder

Circuit:



The Full Adder functions correctly as it can be seen bellow:

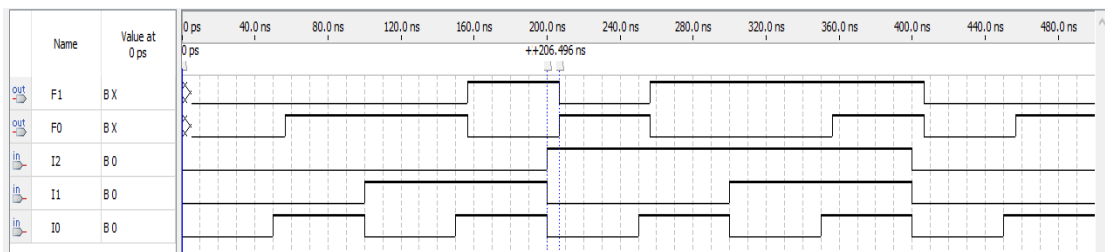




The biggest delay, according to the static analysis, is 6.813ns:

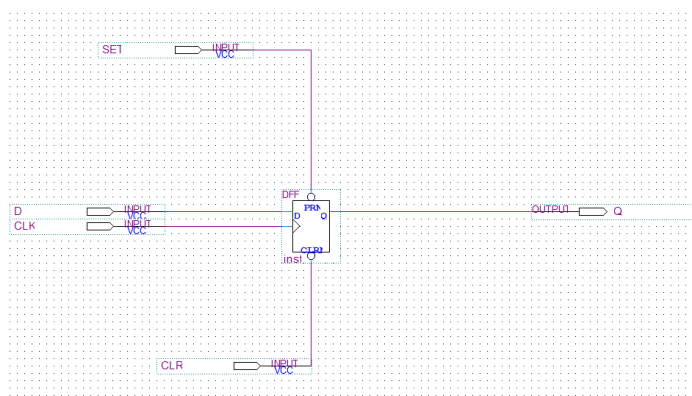
	Input Port	Output Port	RR	RF	FR	FF
1	I0	F0	6.516	6.516	6.516	6.516
2	I0	F1	6.633			6.633
3	I1	F0	6.693	6.693	6.693	6.693
4	I1	F1	6.813			6.813
5	I2	F0	6.381	6.381	6.381	6.381
6	I2	F1	6.503			6.503

The biggest delay, according to the waveform, is 6.496 ns:

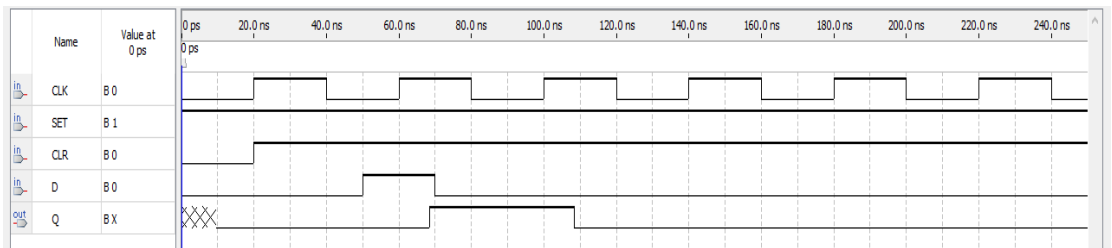
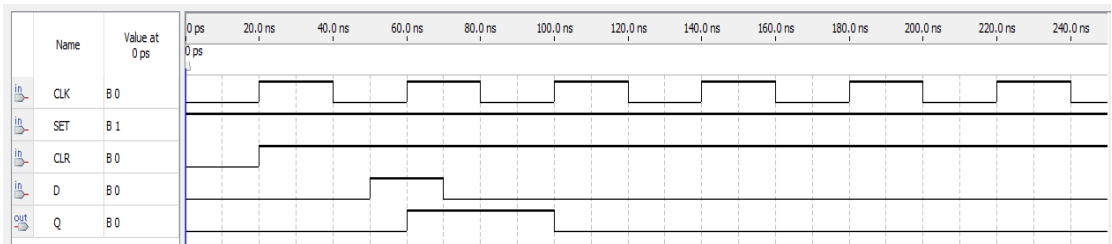


D Flip Flop

Circuit:



The D Flip Flop functions correctly as it can be seen bellow:



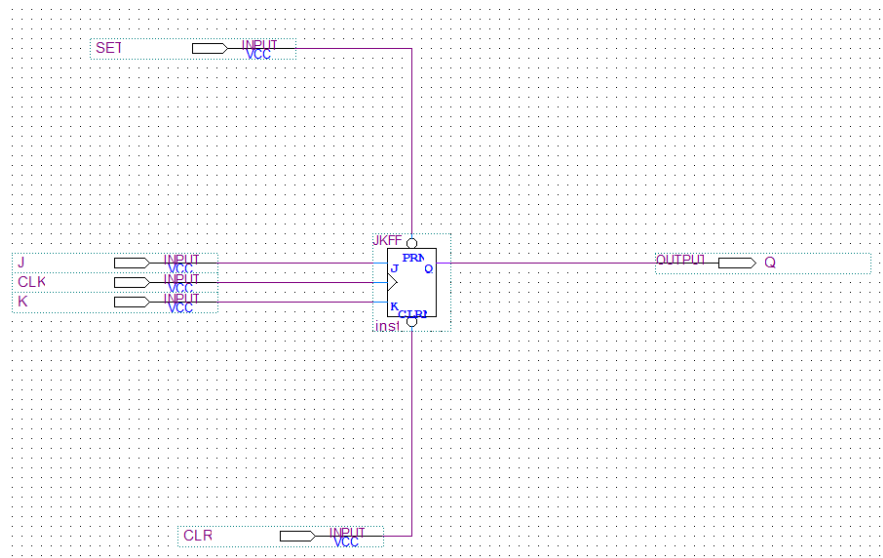
Static Analysis:

	Input Port	Output Port	RR	RF	FR	FF
1	CLR	Q	10.638			10.638
2	SET	Q		11.107	11.107	

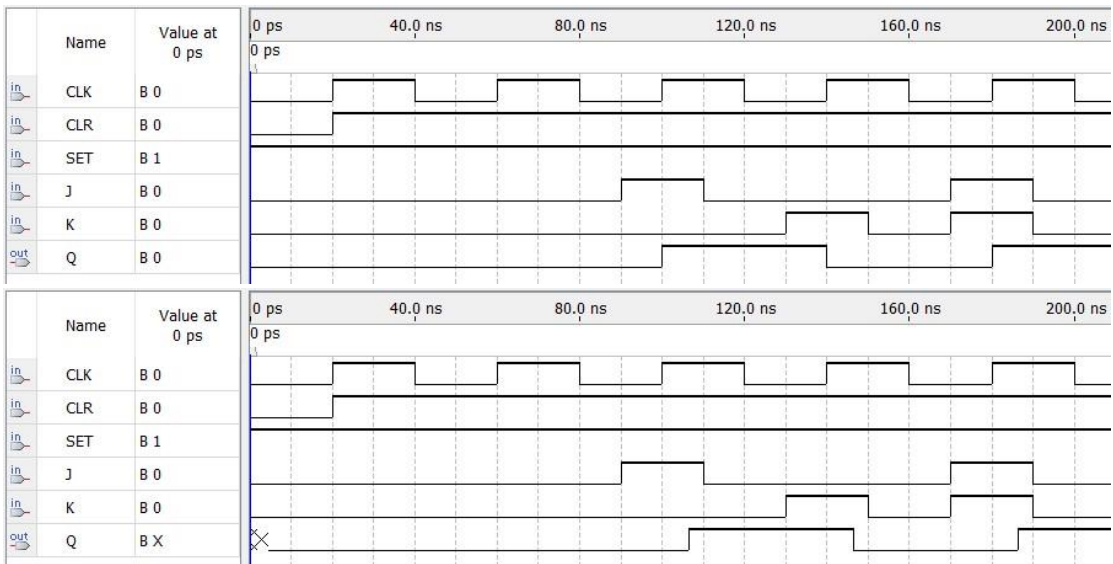
JK Flip Flop & T Flip Flop

JK Flip Flop

Circuit:



The JK Flip Flop functions correctly as it can be seen bellow:

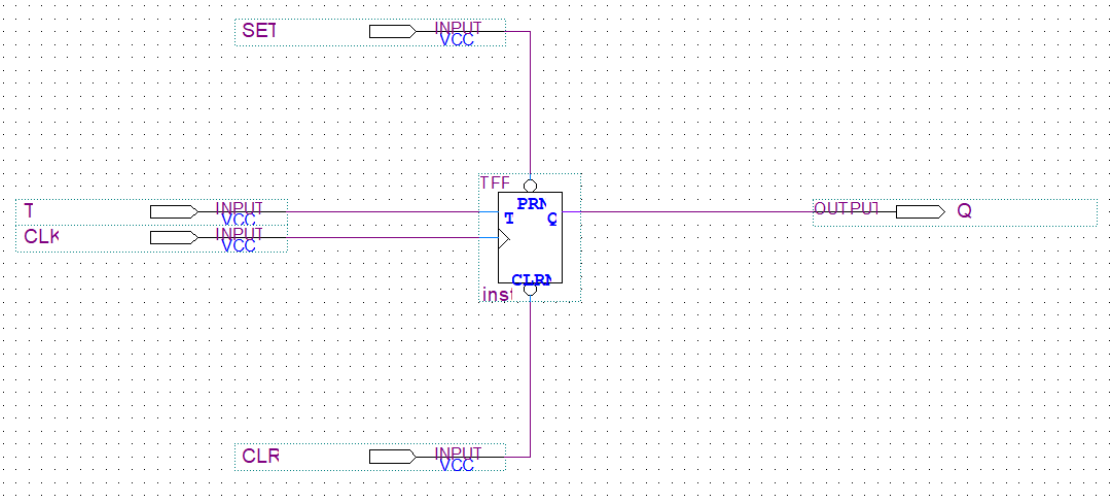


Static Analysis:

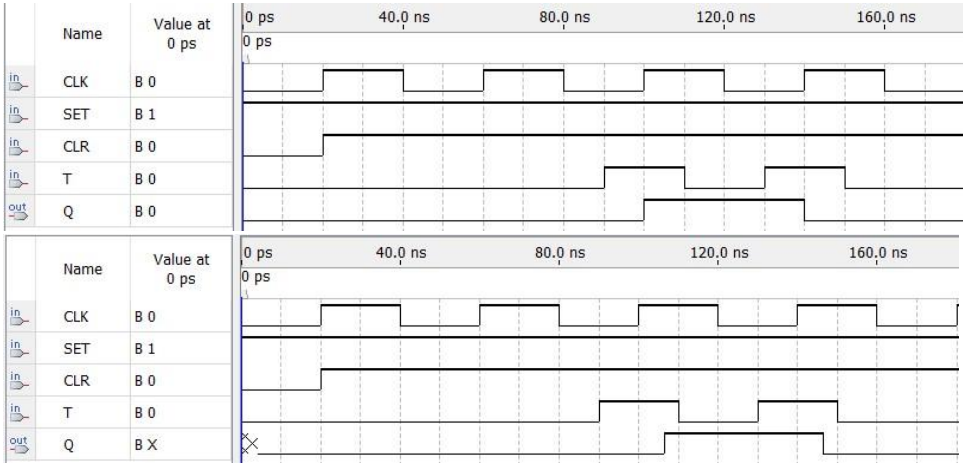
	Input Port	Output Port	RR	RF	FR	FF
1	CLR	Q	11.175			11.175
2	SET	Q		11.303	11.303	

T Flip Flop

Circuit:



The JK Flip Flop functions correctly as it can be seen bellow:

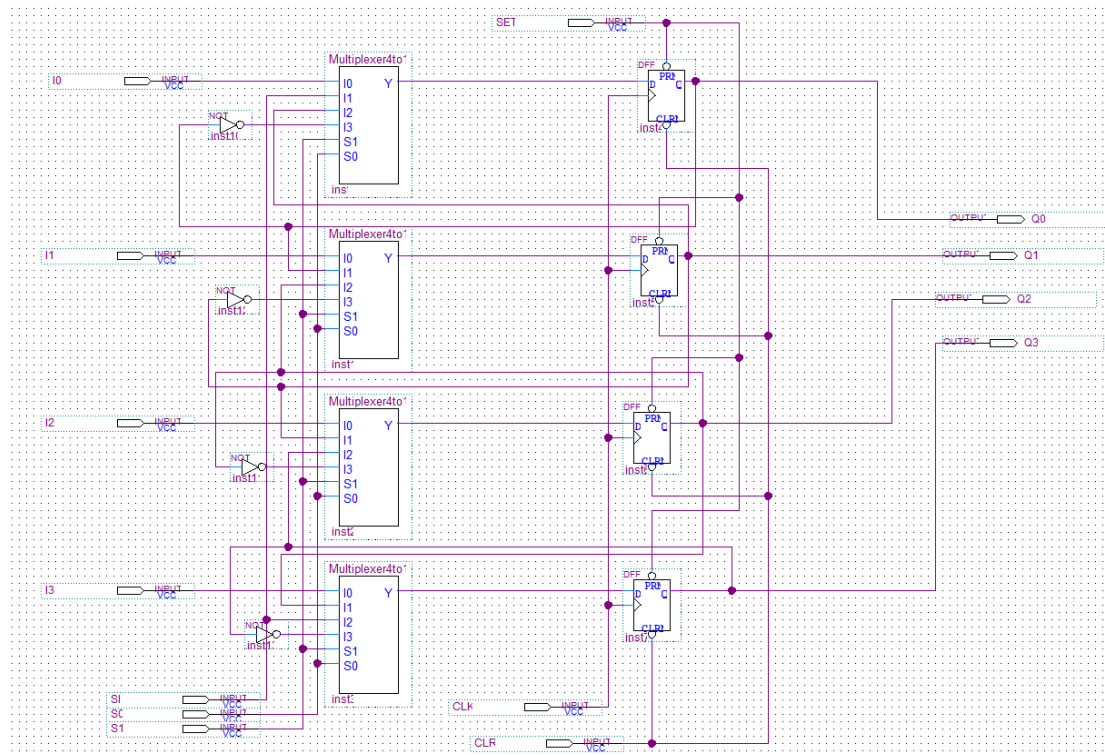


Static Analysis:

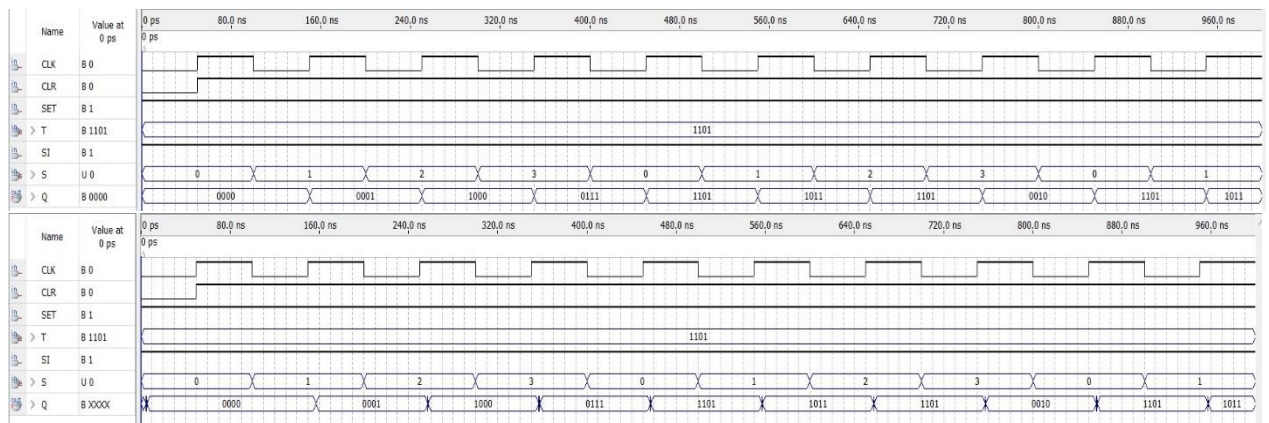
	Input Port	Output Port	RR	RF	FR	FF
1	CLR	Q	11.558			11.558
2	SET	Q		12.003	12.003	

Register

Circuit:



The Register functions correctly as it can be seen bellow:



Static Analysis:

	Input Port	Output Port	RR	RF	FR	FF
1	CLR	Q0	5.434			5.434
2	CLR	Q1	5.749			5.749
3	CLR	Q2	5.933			5.933
4	CLR	Q3	5.637			5.637
5	SET	Q0		5.700	5.700	
6	SET	Q1		5.741	5.741	
7	SET	Q2		5.923	5.923	
8	SET	Q3		5.905	5.905	